Chapter 3 Sec 30-34

The exponential function

We have seen that we can represent a complex number z in the form $z = re^{i\theta}$ but what does it mean to say e^z where z is a complex number?

Let $z = x + iy$ then

$$e^z = e^{x+iy} = e^x e^{iy} = e^x(\cos(y) + i\sin(y))$$ Here we take y in radians

Note if z is real, then $e^z = e^{x+i0} = e^x$ so that we get back our old friend from the real numbers. Further since $e^x \neq 0$ this means that also $e^z \neq 0$

Suppose we assign $w = e^z$, w is itself a complex number and that means we can write it in exponential form $w = \rho e^{i\phi}$ Let's compare the two, Let $z = x + iy$ then.

$\phi = \arg(w)$ $\rho = \|w\|$

$w = e^z \rightarrow \rho e^{i\phi} = e^x e^{iy}$

$\rho = e^x$ $\arg(w) = \arg(e^z) = y + 2n\pi$

So $\rho = e^x$ and $\arg(e^z) = y + 2n\pi$

$z = x + iy$

$|z| = \sqrt{x^2 + y^2}$

What about $|e^z|$?

$|e^z| = |e^x e^{iy}| = |e^x(\cos(y) + i\sin(y))|$

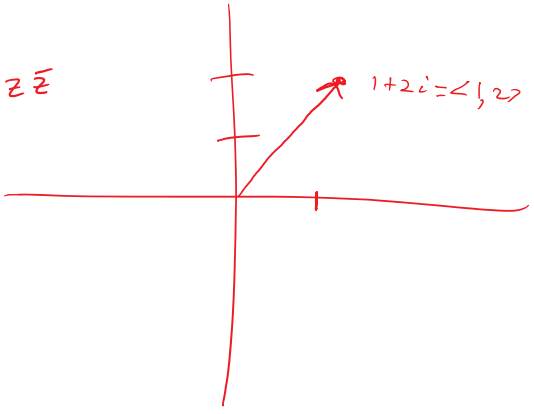
$= |e^x| \, |\cos(y) + i\sin(y)| = e^x \sqrt{\cos^2(y) + \sin^2(y)} = e^x$

We also get some familiar properties of e^z we might remember from calculus.

1. $e^{z_1} e^{z_2} = e^{z_1 + z_2}$
2. $\frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}$
3. $(e^z)^n = e^{nz}$
4. $\frac{d}{dz} e^z = e^z$
5. e^z is entire(analytic everywhere in the complex plane)

$1+2i$

$|z|^2 = z\bar{z}$



Chapter 3 Sec 30-34

Let's prove some of them

$e^{z_1}e^{z_2} = e^{z_1+z_2}$

$e^{x_1+iy_1}e^{x_2+iy_2} = e^{x_1}e^{iy_1}e^{x_2}e^{iy_2}$

$= e^{x_1}e^{x_2}e^{iy_1}e^{iy_2} = e^{(x_1+x_2)}e^{i(y_1+y_2)}$

$= e^{(x_1+x_2)+i(y_1+y_2)} = e^{z_1+z_2}$

$\frac{e^{z_1}}{e^{z_2}} = e^{z_1-z_2}$

$\frac{e^{x_1+iy_1}}{e^{x_2+iy_2}} = \frac{e^{x_1}e^{iy_1}}{e^{x_2}e^{iy_2}} = \frac{e^{x_1}}{e^{x_2}} \cdot \frac{e^{iy_1}}{e^{iy_2}}$

$= e^{x_1-x_2}e^{i(y_1-y_2)} = e^{x_1-x_2+i(y_1-y_2)}$

$= e^{z_1-z_2}$

$z_1 - z_2 = x_1 + iy_1 - (x_2+iy_2)$

$= x_1 - x_2 + i(y_1-y_2)$

Chapter 3 Sec 30-34

Not everything is the same though, in the real numbers $e^x > 0$ Here however e^z can be negative. Just look at $e^{i\pi}$. $= -1$

Another neat fact is that on the complex plane e^z is periodic! Recall a function is called P periodic if $F(z+P) = F(z)$

$e^{i\pi} = \cos\pi + i\sin(i\pi) = -1$

Consider then $F(z) = e^z$ and let's look at $F(z+2\pi i) = e^{z+2\pi i}$ $= e^z e^{i2\pi} = e^z$

Homework example

7. Find all values of z such that
   (a) $e^z = -2$
   (b) $e^z = 1 + \sqrt{3}i$
   (c) $\exp(2z-1) = 1$

$\exp(2z-1) = e^{2z-1}$

$e^{2z-1} = 1$

$e^{2z-1} = e^{i2\pi}$

$e^{2x+2iy-1} = e^{i2\pi}$

$e^{2x-1+2iy} = e^{i2\pi}$

$e^{2x-1}e^{2iy} = e^0 e^{i2\pi}$

$e^{2x-1} = e^0$ $\quad e^{i2y} = e^{i2\pi}$

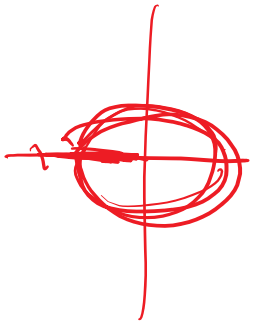
$2x-1 = 0$

$x = 1/2$

$e^{iy} = e^{i\pi}$

$y = \pi + 2n\pi$

$= (2n+1)\pi$ $\quad \pi(1+2n)$



$z = x + iy$

$z = 1/2 + i(2n+1)\pi$

Chapter 3 Sec 30-34

The Logarithmic Function

If we have a complex exponential we would also like the logarithm. That is, we want to solve:

$$e^w = z \text{ for } w$$

Let $w = re^{i\Theta}$ where Θ is the principal argument, the one between $-\pi$ and π then

We can then set the real and imaginary parts equal to each other.

So, $w = \ln(r) + i(\Theta + 2n\pi)$ $n = 0, \pm1, \pm2, \ldots$

Then if we write $\log(z) = \ln(r) + i(\Theta + 2n\pi)$ we could see that $e^{\log(z)} = z$

What about the other way around if we define $\log(z) = \ln(r) + i(\Theta + 2n\pi)$ *is* $\log(e^z) = z$?

Since we get $\log(e^z) = z + 2n\pi i$ we have to be careful and note that $\log(z)$ is a *multiple-valued function*

Chapter 3 Sec 30-34

$Log(e^z) = z$

Note that $\Theta + 2n\pi = \arg(z)$ and $r = |z|$ with this we can write the multiple-valued logarithmic function as

$\log(z) = \ln|z| + i\arg(z)$

We can also define the principal logarithm of $z = re^{i\theta}$ as $Log(z) = \ln(r) + i\Theta$ where Θ is the principal argument of z.

If we do this then $Log(e^z) = \ln|e^z| + iArg(e^z) = \ln(e^x) + i\Theta = x + iy$ when $e^z = e^{x+iy} = e^x e^{iy}$

In this way we have a single valued function. We can then also write if we like:

$\log(z) = \text{Log}(z) + 2n\pi i$

Ok so we have a logarithm, what properties do we get? Unfortunately, as we will see not all of our favorites….

Let z_1, z_2 be any nonzero complex numbers…Then

1. $\log(z_1 z_2) = \log(z_1) + \log(z_2)$

2. $\log\left(\frac{z_1}{z_2}\right) = \log(z_1) - \log(z_2)$

3. $\ln|z_1 z_2| = \ln|z_1| + \ln|z_2|$

4. $z^n = e^{n\log(z)}$

5. $z^{\frac{1}{n}} = e^{\frac{1}{n}\log(z)} = \exp\left[\left(\frac{1}{n}\right)\ln(r) + \left(i\left(\frac{\Theta}{n} + \frac{2k\pi}{n}\right)\right)\right] = r^{\frac{1}{n}}\exp\left[\left(i\left(\frac{\Theta}{n} + \frac{2k\pi}{n}\right)\right)\right]$

Chapter 3 Sec 30-34

We can also say that if $Re(z_1) > 0$ and $Re(z_2) > 0$ then we do have the addition identity for the principal logarithim.

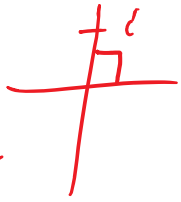
$$Log(z_1 z_2) = Log(z_1) + Log(z_2) \quad if\ Re(z_1) > 0\ \ Re(z_2) > 0$$



The reason for this is that, when both our complex numbers are to the right of the imaginary axis their principal arguments both satisfy $-\frac{\pi}{2} < \Theta < \frac{\pi}{2}$ that means that their sum will satisify , $\pi < \Theta_1 + \Theta_2 < \pi$ so that when added the angle still sits in the principal argument range.

$\frac{-\pi}{2} - \frac{\pi}{2} = -\pi$ $\quad \frac{\pi}{2} + \frac{\pi}{2} = \pi$

You may have noticed that one nice property that we have in real numbers, $\ln(x^n) = n\,ln(x)$ was missing :(. In fact it is in general not true, let's illustrate with a homework problem.



10. (a) Show that $log(i^2) = 2log(i)$ when $\theta \in (\pi/4, 9\pi/4)$
    (b) Show that $log(i^2) \neq 2log(i)$ when $\theta \in (3\pi/4, 11\pi/4)$

$\theta = (-1) + 2n\pi$

$i^2 = (1e^{i\pi/2})^2$

$i^2 (e^{i\pi/2})^2$

a) $\log(i^2) = \ln(|z|) + i(\theta + 2n\pi)$

$= \ln(1) + i(\pi + 2n\pi)$ $\quad i^2 e^{i\pi}$ $\quad n=0$

$= i(\pi + 2n\pi) = i(2n+1)\pi$ $\quad n=0$ $\quad n=0$

$2\log(i) = 2(\ln(1) + i(\pi/2 + 2n\pi))$ $\quad n=1$

$\pi/2 + 2\pi = \frac{\pi}{2} + \frac{4\pi}{2}$

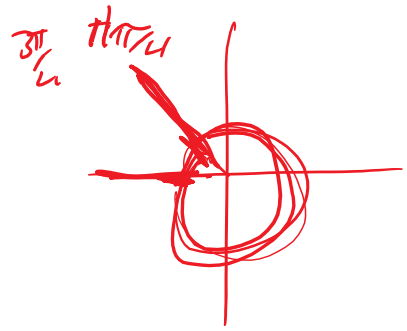
$= 2i(\pi/2 + 2n\pi) = i(\pi + 4n\pi)$ $\quad \frac{5\pi}{2} = \frac{10\pi}{4}$

$= i((4n+1)\pi)$

$i(2n+1)\pi$ $\quad i(4n+1)\pi$

$i\pi$ $\quad i\pi$

$\log(i^2) \stackrel{?}{=} 2\log(i)$ $\quad \theta \in (3\pi/4, 11\pi/4)$



$\log(i^2) = \ln(1) + i(\pi + 2n\pi)$, $2\log(i) = 2(\ln(1) + i(\pi/2 + 2n\pi))$

$n = 0$ $\quad \pi/2$

$n = 0$

$i(\pi)$

$n = 1$ $\quad \pi/2 + 2\pi = \frac{\pi + 4\pi}{2}$

$= 5\pi/2$

$i\frac{10\pi}{4}$

Chapter 3 Sec 30-34

With that pesky multivalued behavior of log taking away our beloved properites we might ask under what conditions might it be single valued…

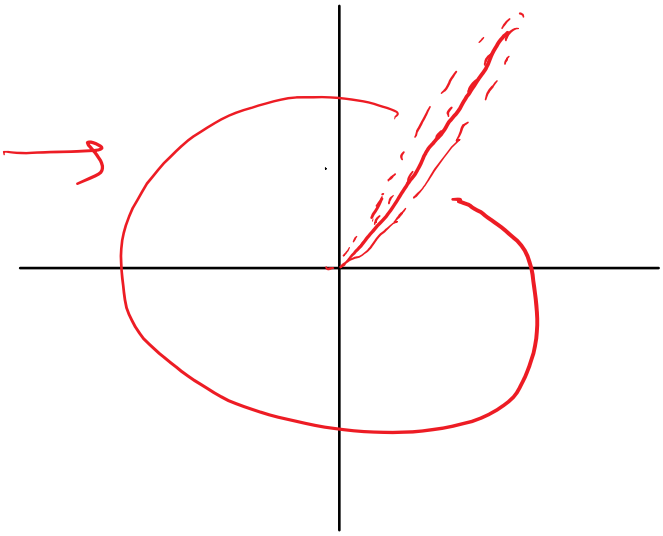
Branches and Derivatives of Logarithms

If $z = re^{i(\Theta)}$ is a nonzero complex number we can get the exact same number by taking any integer multiple of 2π that is $z = re^{i\Theta} = re^{i\theta}$ where $\theta = \Theta + 2n\pi$

With this definition of θ we can write our multi-valued logarithim as $\log(z) = \ln(r) + i\theta$

We can make this multivalued if we restrict theta in the following way. Let α be any real number if we restrict θ to be between α and $\alpha + 2\pi$, $\alpha < \theta < \alpha + 2pi$ then our log(z) function becomes single -valued

Note the strict inequality, that is because our logarithm is not continuous there . Not also that we could choose any alpha, and that as long as we are not on the ray defined at α our log function is single-valued and analtic. We call the α ray a **branch cut**. A branch cut is a curve we introduce into the complex plane to make a multi-valued function single valued on the remaining domain. log(z) on the restricted domain is refered to as a **branch** of the multivalued version of log(z). We will define this more later but let's first show log(z) is indeed analyic on our restricted domain.



$F(z) = \log(z)$ $\quad \arg(z) = \theta + 2n\pi$

Define $F(z) = \ln(r) + i\theta \quad for \quad r > 0\,, \alpha < \theta < \alpha + 2pi$

Then $F(z) = u(r,\theta) + iv(r,\theta)$ with $u(r,\theta) = \ln(r)\ \ v(r,\theta) = \theta$ Checking the polar C-R equations $ru_r = v_\theta, u_\theta = -rv_r$

Since our first partials are continuous everywhere except at the α-ray and they satisify the C-R equations we know that the derivative exists everywhere and infact is equal to

$$\frac{d}{dz}F(z) = \frac{d}{dz}\log(z) = e^{-i\theta}(u_r + iv_r) = e^{-i\theta}\left(\frac{1}{r} + 0i\right) = \frac{1}{re^{i\theta}} = \frac{1}{z} \qquad \text{That is} \qquad \frac{d}{dz}\log(z) = \frac{1}{z}$$

Recall our homework example, they were in fact different branch cuts!

10. (a) Show that $log(i^2) = 2log(i)$ when $\theta \in (\pi/4, 9\pi/4)$
    (b) Show that $log(i^2) \neq 2log(i)$ when $\theta \in (3\pi/4, 11\pi/4)$

This shows us that a function will behave differently on a different branch cut, infact once we make a branch cut we call the new single valued function a Branch of the multi-valued function.

Definition: A Branch of a multi-valued function $f(z)$ is a function $F(z)$ that is analytic in some domain at eac point z of which the value $F(z)$ is one of the values of f.

Definition: A Branch cut is a portion of a line or curve that is introduced in order to define a branch F of a multi-valued function f. Further, points on a branch cut are singular points(not analytic at z_0 but analytic on any neighbor hood of z_0).

Definition: A Branch Point is any point that is common to all branch cuts of $f(z)$.

